OOAD MINI PROJECT

Complaint Management System

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BRIEF DESCRIPTION:

The Complaint Management System is a web-based application that allows users to register and track complaints easily. It stores all complaint data in a centralized database for efficient management and retrieval. The system helps improve transparency by enabling users to monitor the status of their complaints. Overall, it simplifies and streamlines the complaint handling process.

MVC ARCHITECTURE EXPLANATION:

Model

- Represents the data layer.
- Contains entity classes mapped to database tables.
- Example: Complaint, User, etc.
- Uses JPA annotations like @Entity, @Id.

View

- Represents the UI layer.
- Displays data to the user.
- Implemented using: HTML / Thymeleaf

Controller

- Handles user requests and responses.
- Uses annotations like: @Controller or @RestController
- Maps URLs to methods.
- Calls service layer to process logic.

<u>DATABASE DESIGN DETAILS</u>:

1. Admin:

```
mysql> desc admin;
+---------+--------------+------+-----+---------+-------+
| Field   | Type         | Null | Key | Default | Extra |
+---------+--------------+------+-----+---------+-------+
| user_id | int          | NO   | PRI | NULL    |       |
| level   | varchar(255) | YES  |     | NULL    |       |
+---------+--------------+------+-----+---------+-------+
2 rows in set (0.03 sec)
```

2. Complaint:

```
mysql> desc complaint;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| complaint_id | int          | NO   | PRI | NULL    | auto_increment |
| staff_id     | int          | YES  | MUL | NULL    |                |
| user_id      | int          | YES  | MUL | NULL    |                |
| created_at   | datetime(6)  | YES  |     | NULL    |                |
| category     | varchar(255) | YES  |     | NULL    |                |
| description  | varchar(255) | YES  |     | NULL    |                |
| location     | varchar(255) | YES  |     | NULL    |                |
| priority     | varchar(255) | YES  |     | NULL    |                |
| status       | varchar(255) | YES  |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
9 rows in set (0.02 sec)
```

3. Complaint_history:

```
mysql> desc complaint_history;
+------------------------+--------------+------+-----+---------+----------------+
| Field                  | Type         | Null | Key | Default | Extra          |
+------------------------+--------------+------+-----+---------+----------------+
| complaint_complaint_id | int          | YES  | MUL | NULL    |                |
| log_id                 | int          | NO   | PRI | NULL    | auto_increment |
| changed_at             | datetime(6)  | YES  |     | NULL    |                |
| new_status             | varchar(255) | YES  |     | NULL    |                |
| old_status             | varchar(255) | YES  |     | NULL    |                |
| remarks                | varchar(255) | YES  |     | NULL    |                |
+------------------------+--------------+------+-----+---------+----------------+
6 rows in set (0.01 sec)
```

4. Feedback:

```
mysql> desc feedback;
+------------------------+--------------+------+-----+---------+----------------+
| Field                  | Type         | Null | Key | Default | Extra          |
+------------------------+--------------+------+-----+---------+----------------+
| complaint_complaint_id | int          | YES  | MUL | NULL    |                |
| feedback_id            | int          | NO   | PRI | NULL    | auto_increment |
| rating                 | int          | NO   |     | NULL    |                |
| comments               | varchar(255) | YES  |     | NULL    |                |
+------------------------+--------------+------+-----+---------+----------------+
4 rows in set (0.02 sec)
```

5. Payment:

```
mysql> desc payment;
+-----------------------+--------------+------+-----+---------+----------------+
| Field                 | Type         | Null | Key | Default | Extra          |
+-----------------------+--------------+------+-----+---------+----------------+
| amount                | double       | NO   |     | NULL    |                |
| complaint_complaint_id | int          | YES  | UNI | NULL    |                |
| payment_id            | int          | NO   | PRI | NULL    | auto_increment |
| transaction_time      | datetime(6)  | YES  |     | NULL    |                |
| payment_mode          | varchar(255) | YES  |     | NULL    |                |
| payment_status        | varchar(255) | YES  |     | NULL    |                |
+-----------------------+--------------+------+-----+---------+----------------+
6 rows in set (0.01 sec)
```

6. Report:

```
mysql> desc report;
+----------------------+------+------+-----+---------+----------------+
| Field                | Type | Null | Key | Default | Extra          |
+----------------------+------+------+-----+---------+----------------+
| generated_by_user_id | int  | YES  | MUL | NULL    |                |
| month_year           | date | YES  |     | NULL    |                |
| pending_complaints   | int  | NO   |     | NULL    |                |
| report_id            | int  | NO   | PRI | NULL    | auto_increment |
| resolved_complaints  | int  | NO   |     | NULL    |                |
| total_complaints     | int  | NO   |     | NULL    |                |
+----------------------+------+------+-----+---------+----------------+
6 rows in set (0.01 sec)
```

7. Resident:

```
mysql> desc resident;
+---------+--------+------+-----+---------+-------+
| Field   | Type   | Null | Key | Default | Extra |
+---------+--------+------+-----+---------+-------+
| balance | double | NO   |     | NULL    |       |
| user_id | int    | NO   | PRI | NULL    |       |
+---------+--------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

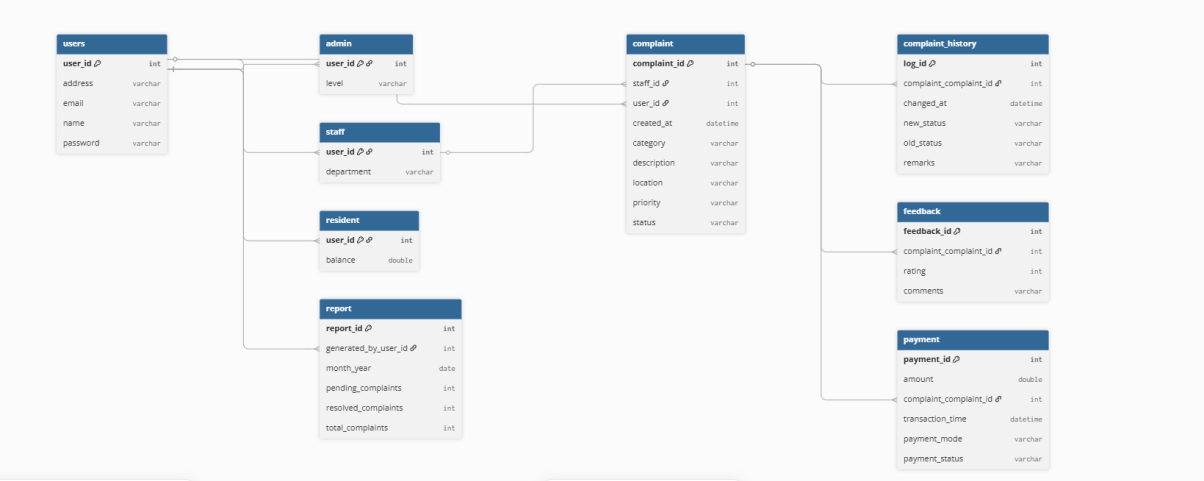
8. Staff:

```
mysql> desc staff;
+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| user_id    | int          | NO   | PRI | NULL    |       |
| department | varchar(255) | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
2 rows in set (0.01 sec)
```

9. Users:

```
mysql> desc users;
+----------+--------------+------+-----+---------+----------------+
| Field    | Type         | Null | Key | Default | Extra          |
+----------+--------------+------+-----+---------+----------------+
| user_id  | int          | NO   | PRI | NULL    | auto_increment |
| address  | varchar(255) | YES  |     | NULL    |                |
| email    | varchar(255) | YES  |     | NULL    |                |
| name     | varchar(255) | YES  |     | NULL    |                |
| password | varchar(255) | YES  |     | NULL    |                |
+----------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

ER DIAGRAM:



SPRINGBOOT CODE:

Controller:

1.ComplaintController.java:

complainttracker_final > complainttracker > src > main > java > com > project > complainttracker > controller > J ComplaintController.java > Language Supp

```java
package com.project.complainttracker.controller;

import com.project.complainttracker.model.Complaint;
import com.project.complainttracker.model.User;
import com.project.complainttracker.service.ComplaintService;
import com.project.complainttracker.service.UserService;

import jakarta.servlet.http.HttpSession;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.*;

@Controller
public class ComplaintController {

    @Autowired
    private ComplaintService service;

    // Show form
    @GetMapping("/user/add-complaint")
    public String showForm() {
        return "add-complaint";
    }

    // Handle form submission
    @PostMapping("/user/add-complaint")
    public String addComplaint(@RequestParam String description,
                               HttpSession session) {
```

```java

        User user = (User) session.getAttribute("loggedUser");

        Complaint complaint = new Complaint();
        complaint.setDescription(description);
        complaint.setUser(user);

        service.addComplaint(complaint);

        return "user-dashboard";
    }

    @GetMapping("/user/view-complaints")
    public String viewComplaints(Model model, HttpSession session) {

        User user = (User) session.getAttribute("loggedUser");

        model.addAttribute("complaints", service.getUserComplaints(user));

        return "view-complaints";
    }

    @GetMapping("/admin/assign")
    public String showAssignPage(Model model) {

       model.addAttribute("complaints", service.getAllComplaints());
       model.addAttribute("staffList", userService.getStaffUsers());
```

```java

        return "assign-complaint";
    }

    @Autowired
    private UserService userService;

    @PostMapping("/admin/assign")
    public String assignComplaint(@RequestParam int complaintId,
                                  @RequestParam int staffId) {

        User staff = userService.getStaffUsers()
            .stream()
            .filter(s -> s.getUserId() == staffId)
            .findFirst()
            .orElse(null);

        service.assignStaff(complaintId, staff);

        return "admin-dashboard";
    }

    @GetMapping("/admin/complaints")
    public String viewAllComplaints(Model model) {

        model.addAttribute("complaints", service.getAllComplaints());

```

```java
        return "admin-complaints";
    }

    @GetMapping("/admin/delete/{id}")
    public String deleteComplaint(@PathVariable int id) {

        service.deleteComplaint(id);

        return "redirect:/admin/complaints";
    }

    @GetMapping("/admin/edit/{id}")
    public String showEditForm(@PathVariable int id, Model model) {

        Complaint complaint = service.getAllComplaints()
            .stream()
            .filter(c -> c.getComplaintId() == id)
            .findFirst()
            .orElse(null);

        model.addAttribute("complaint", complaint);

        return "edit-complaint";
    }

    @PostMapping("/admin/edit")
    public String updateComplaint(@RequestParam int complaintId,
```

```java
                                @RequestParam String description,
                                @RequestParam String status) {

        Complaint complaint = service.getAllComplaints()
            .stream()
            .filter(c -> c.getComplaintId() == complaintId)
            .findFirst()
            .orElse(null);

        if (complaint != null) {
        complaint.setDescription(description);
        complaint.setStatus(status);
        service.addComplaint(complaint);
        }

        return "redirect:/admin/complaints";
    }

    @GetMapping("/staff/complaints")
    public String viewStaffComplaints(Model model, HttpSession session) {

        User staff = (User) session.getAttribute("loggedUser");

        model.addAttribute("complaints", service.getStaffComplaints(staff));

        return "staff-complaints";
    }
```

```java

}
```

2.LoginController.java:

complainttracker_final > complainttracker > src > main > java > com > project > complainttracker > controller > LoginController.java > Language Support

```java
package com.project.complainttracker.controller;

import com.project.complainttracker.model.User;
import com.project.complainttracker.service.UserService;
import jakarta.servlet.http.HttpSession;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.*;

@Controller
public class LoginController {

    @Autowired
    private UserService service;

    // Show login page
    @GetMapping("/")
    public String showLoginPage() {
        return "login";
    }

    // Handle login
    @PostMapping("/login")
    public String login(@RequestParam String email,
                        @RequestParam String password,
                        HttpSession session) {

        User user = service.login(email, password);

```

```java
        if(user != null) {

            session.setAttribute("loggedUser", user);

            if(user.getRole().equals("ADMIN")) {
                return "admin-dashboard";
            }
            else if(user.getRole().equals("STAFF")) {
                return "staff-dashboard";
            }
            else {
                return "user-dashboard";
            }
        }

        return "login";
    }

    @GetMapping("/logout")
    public String logout(HttpSession session) {
        session.invalidate(); // clears session
        return "redirect:/";
    }

    @GetMapping("/user-dashboard")
    public String userDashboard() {
        return "user-dashboard";
```

```java
    }

    @GetMapping("/admin-dashboard")
    public String adminDashboard() {
        return "admin-dashboard";
    }

    @GetMapping("/staff-dashboard")
    public String staffDashboard() {
        return "staff-dashboard";
    }
}
```

Model:

1.Complaint.java:

complainttracker_final > complainttracker > src > main > java > com > project > complainttracker > model > Complaint.java > { } com.project.complainttra

```java
package com.project.complainttracker.model;

import jakarta.persistence.*;
import java.time.LocalDateTime;

@Entity
public class Complaint {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private int complaintId;

    private String description;
    private String status;

    private LocalDateTime createdAt;

    // USER who created complaint
    @ManyToOne
    @JoinColumn(name = "user_id")
    private User user;

    // STAFF assigned
    @ManyToOne
    @JoinColumn(name = "staff_id")
    private User staff;

    // Getters and Setters

```

```java
    public int getComplaintId() {
        return complaintId;
    }

    public void setComplaintId(int complaintId) {
        this.complaintId = complaintId;
    }

    public String getDescription() {
        return description;
    }

    public void setDescription(String description) {
        this.description = description;
    }

    public String getStatus() {
        return status;
    }

    public void setStatus(String status) {
        this.status = status;
    }

    public LocalDateTime getCreatedAt() {
        return createdAt;
    }

```

```java
    public void setCreatedAt(LocalDateTime createdAt) {
        this.createdAt = createdAt;
    }

    public User getUser() {
        return user;
    }

    public void setUser(User user) {
        this.user = user;
    }

    public User getStaff() {
        return staff;
    }

    public void setStaff(User staff) {
        this.staff = staff;
    }
}
```

2.User.java:

complainttracker_final > complainttracker > src > main > java > com > project > complainttracker > model > User.java > { } com.project.complainttracker.m

```java
package com.project.complainttracker.model;

import jakarta.persistence.*;

@Entity
@Table(name = "users")
public class User {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private int userId;

    private String name;
    private String email;
    private String password;

    private String role; // USER, STAFF, ADMIN

    // Getters and Setters

    public int getUserId() {
        return userId;
    }

    public void setUserId(int userId) {
        this.userId = userId;
    }

    public String getName() {
```

```java
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }

    public String getPassword() {
        return password;
    }

    public void setPassword(String password) {
        this.password = password;
    }

    public String getRole() {
        return role;
    }

```

```java
    public void setRole(String role) {
        this.role = role;
    }
}
```

Repository:

1.ComplaintRepository.java:

complainttracker_final > complainttracker > src > main > java > com > project > complainttracker > repository > ComplaintRepository.java > Language Sup

```java
package com.project.complainttracker.repository;

import com.project.complainttracker.model.Complaint;
import com.project.complainttracker.model.User;
import org.springframework.data.jpa.repository.JpaRepository;

import java.util.List;

public interface ComplaintRepository extends JpaRepository<Complaint, Integer> {

    Show AOT-generated Implementation, Query, etc...
    List<Complaint> findByUser(User user);

    Show AOT-generated Implementation, Query, etc...
    List<Complaint> findByStaff(User staff);
}
```

2.UserRepository.java:

n > project > complainttracker > repository > UserRepository.java > Language Support for Java(TM) by Red Hat > { } com.project.complainttracker.repository

```java
package com.project.complainttracker.repository;

import com.project.complainttracker.model.User;
import org.springframework.data.jpa.repository.JpaRepository;
import java.util.List;

public interface UserRepository extends JpaRepository<User, Integer> {

    Show AOT-generated Implementation, Query, etc...
    User findByEmail(String email);
    Show AOT-generated Implementation, Query, etc...
    List<User> findByRole(String role);
}

```

Service:

1.ComplaintService.java:

ComplainttrackerApplication.java | UserRepository.java | ComplaintService.java | ComplaintRepository.java

complainttracker_final > complainttracker > src > main > java > com > project > complainttracker > service > ComplaintService.java > Language Support fo

```java
package com.project.complainttracker.service;

import com.project.complainttracker.model.Complaint;
import com.project.complainttracker.model.User;
import com.project.complainttracker.repository.ComplaintRepository;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import java.time.LocalDateTime;
import java.util.List;

@Service
public class ComplaintService {

    @Autowired
    private ComplaintRepository repo;

    public Complaint addComplaint(Complaint complaint) {
        complaint.setStatus(status: "Pending");
        complaint.setCreatedAt(LocalDateTime.now());
        return repo.save(complaint);
    }

    public List<Complaint> getAllComplaints() {
        return repo.findAll();
    }

    public List<Complaint> getUserComplaints(User user) {
        return repo.findByUser(user);
```

```java
    }

    public List<Complaint> getStaffComplaints(User staff) {
        return repo.findByStaff(staff);
    }

    public void deleteComplaint(int id) {
        repo.deleteById(id);
    }

    public void assignStaff(int complaintId, User staff) {
        Complaint complaint = repo.findById(complaintId).orElse(null);

        if (complaint != null) {
            complaint.setStaff(staff);
            complaint.setStatus(status: "In Progress");
            repo.save(complaint);
        }
    }
}
```

2.UserService.java:

complainttracker_final > complainttracker > src > main > java > com > project > complainttracker > service > UserService.java > Language Support for Java

```java
package com.project.complainttracker.service;

import com.project.complainttracker.model.User;
import com.project.complainttracker.repository.UserRepository;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import java.util.List;


@Service
public class UserService {

    @Autowired
    private UserRepository repo;

    public User login(String email, String password) {
        User user = repo.findByEmail(email);

        if(user != null && user.getPassword().equals(password)) {
            return user;
        }

        return null;
    }

    public User saveUser(User user) {
        return repo.save(user);
    }

```

```java
    public List<User> getStaffUsers() {
        return repo.findByRole(role: "STAFF");
    }
}
```

ComplainttrackerApplication.java:

complainttracker_final > complainttracker > src > main > java > com > project > complainttracker > ComplainttrackerApplication.java > Language Support

```java
package com.project.complainttracker;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class ComplainttrackerApplication {

    Run | Debug
    public static void main(String[] args) {
        SpringApplication.run(ComplainttrackerApplication.class, args);
    }

}
```

ComplainttrackerApplicationTests.java:

complainttracker_final > complainttracker > src > test > java > com > project > complainttracker > ComplainttrackerApplicationTests.java > { } com.project.

```java
package com.project.complainttracker;

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;

@SpringBootTest
class ComplainttrackerApplicationTests {

    @Test
    void contextLoads() {
    }

}

```

1.Add-complaint.html:

complainttracker_final > complainttracker > src > main > resources > templates > add-complaint.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Add Complaint</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .form-container {
            background-color: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 350px;
            text-align: center;
        }

        h2 {
            margin-bottom: 20px;
        }

        input {
```

```
            width: 90%;
            padding: 8px;
            margin: 10px 0;
            border-radius: 5px;
            border: 1px solid #ccc;
        }

        button {
            width: 100%;
            padding: 10px;
            background-color: #007bff;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
            background-color: #0056b3;
        }

        .back-btn {
            display: inline-block;
            margin-top: 15px;
            text-decoration: none;
```

```
            color: #555;
        }

        .back-btn:hover {
            text-decoration: underline;
        }
    </style>
</head>

<body>

<div class="form-container">
    <h2>Add Complaint</h2>

    <form action="/user/add-complaint" method="post">
        <input type="text" name="description" placeholder="Enter complaint" required>

        <button type="submit">Submit</button>
    </form>

    <a href="/user-dashboard" class="back-btn">← Back</a>
</div>

</body>
</html>
```

2.admin-complaints.html:

complainttracker_final > complainttracker > src > main > resources > templates > admin-complaints.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Admin Complaints</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .table-container {
            background-color: white;
            padding: 25px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 600px;
            text-align: center;
        }

        h2 {
            margin-bottom: 20px;
        }

        table {
```

```
            width: 100%;
            border-collapse: collapse;
        }

        th {
            background-color: #28a745; /* admin green */
            color: white;
            padding: 10px;
        }

        td {
            padding: 10px;
            border-bottom: 1px solid #ddd;
        }

        tr:hover {
            background-color: #f1f1f1;
        }

        .action-btn {
            padding: 5px 10px;
            text-decoration: none;
            border-radius: 5px;
            color: white;
            margin: 2px;
```

```
        }

        .edit-btn {
            background-color: #007bff;
        }

        .edit-btn:hover {
            background-color: #0056b3;
        }

        .delete-btn {
            background-color: #dc3545;
        }

        .delete-btn:hover {
            background-color: #a71d2a;
        }

        .back-btn {
            display: inline-block;
            margin-top: 15px;
            text-decoration: none;
            color: #555;
        }

```

```
        .back-btn:hover {
            text-decoration: underline;
        }
    </style>
</head>

<body>

<div class="table-container">
    <h2>All Complaints</h2>

    <table>
        <tr>
            <th>ID</th>
            <th>Description</th>
            <th>Status</th>
            <th>Actions</th>
        </tr>

        <tr th:each="c : ${complaints}">
            <td th:text="${c.complaintId}"></td>
            <td th:text="${c.description}"></td>
            <td th:text="${c.status}"></td>

            <td>
```

```html
                <a th:href="@{'/admin/edit/' + ${c.complaintId}}" class="action-btn edit-btn">Edit</a
                <a th:href="@{'/admin/delete/' + ${c.complaintId}}" class="action-btn delete-btn">Del
            </td>
        </tr>
    </table>

    <a href="/admin-dashboard" class="back-btn">← Back</a>
</div>

</body>
</html>
```

3.admin-dashboard.html:

complainttracker_final > complainttracker > src > main > resources > templates > admin-dashboard.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Admin Dashboard</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .dashboard-container {
            background-color: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 350px;
            text-align: center;
        }

        h2 {
            margin-bottom: 10px;
        }

        p {
```

```
            margin-bottom: 20px;
            color: #555;
        }

        .btn {
            display: block;
            margin: 10px 0;
            padding: 10px;
            background-color: #28a745; /* green for admin */
            color: white;
            text-decoration: none;
            border-radius: 5px;
        }

        .btn:hover {
            background-color: #1e7e34;
        }

        .logout {
            background-color: #dc3545;
        }

        .logout:hover {
            background-color: #a71d2a;
        }
```

```
    </style>
</head>

<body>

<div class="dashboard-container">
    <h2>Welcome Admin</h2>
    <p>Manage and control all complaints</p>

    <a href="/admin/assign" class="btn">Assign Complaints</a>
    <a href="/admin/complaints" class="btn">Manage Complaints</a>
    <a href="/logout" class="btn logout">Logout</a>
</div>

</body>
</html>
```

4.assign-complaint.html:

complainttracker_final > complainttracker > src > main > resources > templates > assign-complaint.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Assign Complaint</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .form-container {
            background-color: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 350px;
            text-align: center;
        }

        h2 {
            margin-bottom: 20px;
        }

        label {
```

```
            display: block;
            margin-top: 10px;
            margin-bottom: 5px;
            text-align: left;
        }

        select {
            width: 100%;
            padding: 8px;
            border-radius: 5px;
            border: 1px solid #ccc;
        }

        button {
            margin-top: 15px;
            width: 100%;
            padding: 10px;
            background-color: #28a745;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
```

```
            background-color: #1e7e34;
        }

        .back-btn {
            display: inline-block;
            margin-top: 15px;
            text-decoration: none;
            color: #555;
        }

        .back-btn:hover {
            text-decoration: underline;
        }
    </style>
</head>

<body>

<div class="form-container">
    <h2>Assign Complaint</h2>

    <form action="/admin/assign" method="post">

        <label>Complaint</label>
        <select name="complaintId">
```

```
            <option th:each="c : ${complaints}"
                    th:value="${c.complaintId}"
                    th:text="${c.description}">
            </option>
        </select>

        <label>Staff</label>
        <select name="staffId">
            <option th:each="s : ${staffList}"
                    th:value="${s.userId}"
                    th:text="${s.name}">
            </option>
        </select>

        <button type="submit">Assign</button>
    </form>

    <a href="/admin-dashboard" class="back-btn">← Back</a>
</div>

</body>
</html>
```

5.edit-complaint.html:

complainttracker_final > complainttracker > src > main > resources > templates > edit-complaint.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Edit Complaint</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .form-container {
            background-color: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 350px;
            text-align: center;
        }

        h2 {
            margin-bottom: 20px;
        }

        label {
```

```
            display: block;
            text-align: left;
            margin-top: 10px;
        }

        input {
            width: 100%;
            padding: 8px;
            margin-top: 5px;
            border-radius: 5px;
            border: 1px solid #ccc;
        }

        button {
            margin-top: 15px;
            width: 100%;
            padding: 10px;
            background-color: #28a745; /* admin green */
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
```

```
            background-color: #1e7e34;
        }

        .back-btn {
            display: inline-block;
            margin-top: 15px;
            text-decoration: none;
            color: #555;
        }

        .back-btn:hover {
            text-decoration: underline;
        }
    </style>
</head>

<body>

<div class="form-container">
    <h2>Edit Complaint</h2>

    <form action="/admin/edit" method="post">

        <input type="hidden" name="complaintId" th:value="${complaint.complaintId}">

```

```
        <label>Description</label>
        <input type="text" name="description" th:value="${complaint.description}" required>

        <label>Status</label>
        <input type="text" name="status" th:value="${complaint.status}" required>

        <button type="submit">Update</button>
    </form>

    <a href="/admin-dashboard" class="back-btn">← Back</a>
</div>

</body>
</html>
```

6.Login.html:

complainttracker_final > complainttracker > src > main > resources > templates > login.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Login</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .login-container {
            background-color: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 300px;
            text-align: center;
        }

        h2 {
            margin-bottom: 20px;
        }

        input {
            width: 90%;
            padding: 8px;
            margin: 10px 0;
            border-radius: 5px;
            border: 1px solid #ccc;
        }

        button {
            width: 100%;
            padding: 10px;
            background-color: #007bff;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
            background-color: #0056b3;
        }
    </style>
</head>

<body>

```

```html
<div class="login-container">
    <h2>Login</h2>

    <form action="/login" method="post">
        <input type="text" name="email" placeholder="Email" required>
        <input type="password" name="password" placeholder="Password" required>

        <button type="submit">Login</button>
    </form>
</div>

</body>
</html>
```

7.Staff-complaints.html:

complainttracker_final > complainttracker > src > main > resources > templates > staff-complaints.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Staff Complaints</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .table-container {
            background-color: white;
            padding: 25px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 500px;
            text-align: center;
        }

        h2 {
            margin-bottom: 20px;
        }

        table {
```

```
30            width: 100%;
31            border-collapse: collapse;
32        }
33
34        th {
35            background-color: #17a2b8; /* staff color */
36            color: white;
37            padding: 10px;
38        }
39
40        td {
41            padding: 10px;
42            border-bottom: 1px solid #ddd;
43        }
44
45        tr:hover {
46            background-color: #f1f1f1;
47        }
48
49        .back-btn {
50            display: inline-block;
51            margin-top: 15px;
52            text-decoration: none;
53            color: #555;
54        }
```

```
55
56            .back-btn:hover {
57                text-decoration: underline;
58            }
59        </style>
60    </head>
61
62    <body>
63
64    <div class="table-container">
65        <h2>Assigned Complaints</h2>
66
67        <table>
68            <tr>
69                <th>ID</th>
70                <th>Description</th>
71                <th>Status</th>
72            </tr>
73
74            <tr th:each="c : ${complaints}">
75                <td th:text="${c.complaintId}"></td>
76                <td th:text="${c.description}"></td>
77                <td th:text="${c.status}"></td>
78            </tr>
79        </table>
80
```

```html
        <a href="/staff-dashboard" class="back-btn">← Back</a>
    </div>

    </body>
    </html>
```

8.Staff-dashboard.java:

complainttracker_final > complainttracker > src > main > resources > templates > staff-dashboard.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Staff Dashboard</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .dashboard-container {
            background-color: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 350px;
            text-align: center;
        }

        h2 {
            margin-bottom: 10px;
        }

        p {
```

```
            margin-bottom: 20px;
            color: #555;
        }

        .btn {
            display: block;
            margin: 10px 0;
            padding: 10px;
            background-color: #17a2b8; /* teal for staff */
            color: white;
            text-decoration: none;
            border-radius: 5px;
        }

        .btn:hover {
            background-color: #117a8b;
        }

        .logout {
            background-color: #dc3545;
        }

        .logout:hover {
            background-color: #a71d2a;
        }
```

```
    </style>
</head>

<body>

<div class="dashboard-container">
    <h2>Welcome Staff</h2>
    <p>View and manage assigned complaints</p>

    <a href="/staff/complaints" class="btn">View Assigned Complaints</a>
    <a href="/logout" class="btn logout">Logout</a>
</div>

</body>
</html>
```

9.User-dashboard.html:

complainttracker_final > complainttracker > src > main > resources > templates > user-dashboard.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>User Dashboard</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .dashboard-container {
            background-color: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 350px;
            text-align: center;
        }

        h2 {
            margin-bottom: 10px;
        }

        p {
            margin-bottom: 20px;
            color: #555;
        }

        .btn {
            display: block;
            margin: 10px 0;
            padding: 10px;
            background-color: #007bff;
            color: white;
            text-decoration: none;
            border-radius: 5px;
        }

        .btn:hover {
            background-color: #0056b3;
        }

        .logout {
            background-color: #dc3545;
        }

        .logout:hover {
            background-color: #a71d2a;
        }
```

```html
        </style>
    </head>

    <body>

    <div class="dashboard-container">
        <h2>Welcome User</h2>
        <p>Manage your complaints easily</p>

        <a href="/user/add-complaint" class="btn">Add Complaint</a>
        <a href="/user/view-complaints" class="btn">View My Complaints</a>
        <a href="/logout" class="btn logout">Logout</a>
    </div>

    </body>
    </html>
```

10.View-complaints.html:

complainttracker_final > complainttracker > src > main > resources > templates > view-complaints.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>My Complaints</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .table-container {
            background-color: white;
            padding: 25px;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
            width: 500px;
            text-align: center;
        }

        h2 {
            margin-bottom: 20px;
        }

        table {
```

```
        width: 100%;
        border-collapse: collapse;
    }

    th {
        background-color: #007bff;
        color: white;
        padding: 10px;
    }

    td {
        padding: 10px;
        border-bottom: 1px solid #ddd;
    }

    tr:hover {
        background-color: #f1f1f1;
    }

    .back-btn {
        display: inline-block;
        margin-top: 15px;
        text-decoration: none;
        color: #555;
    }
```

```
        .back-btn:hover {
            text-decoration: underline;
        }
    </style>
</head>

<body>

<div class="table-container">
    <h2>My Complaints</h2>

    <table>
        <tr>
            <th>ID</th>
            <th>Description</th>
            <th>Status</th>
        </tr>

        <tr th:each="c : ${complaints}">
            <td th:text="${c.complaintId}"></td>
            <td th:text="${c.description}"></td>
            <td th:text="${c.status}"></td>
        </tr>
    </table>

```

```
    <a href="/user-dashboard" class="back-btn">← Back</a>
</div>

</body>
</html>
```

Application.properties:

complainttracker_final > complainttracker > src > main > resources > application.properties

```
spring.application.name=complainttracker
spring.datasource.url=jdbc:mysql://localhost:3306/complaint_db
spring.datasource.username=root
spring.datasource.password=Neha2805

spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
```

GRASP PRINCIPLES USED:

1. Controller - ComplaintController class

   Methods like:

   Add Complaint

   View Complaints

2. Information Expert - Complaint model class (stores complaint data)
   ComplaintRepository (handles database operations)

3. Low Coupling - Controller (interacts only with Service)
   Service (interacts only with Repository)

4. High Cohesion - ComplaintController (handles requests)
   ComplaintService (processes logic)
   ComplaintRepository (manages database)

DESIGN PATTERNS USED:

1. Model-View-Controller (MVC) Pattern - Entire project structure is based on MVC
   ComplaintController, Complaint, and frontend/response handling

2. Repository Pattern - ComplaintRepository interface
   Extends JpaRepository to perform CRUD operations

3. Dependency Injection Pattern - Injecting ComplaintService into ComplaintController
   Injecting ComplaintRepository into ComplaintService

4. Service Layer Pattern - ComplaintService class
   Handles operations like adding, updating, and processing complaints